BBA 2421
Business Mathematics

Lecture 5: Annuities, Perpetuities and Amortization

Department of Economics

Annuities and Perpetuities

Introduction

- We have observed that the future value and present value concepts can be used to answer a number of problems concerning the time value of money.
- However, the human effort involved in the computations can be excessive.
- For example, imagine yourself calculating the present value on a 30-year monthly mortgage.
- Since the mortgage has 600 (30 x 12) payments, a lot of time may be required to perform a conceptually simple task.

Introduction cont…

- Owing to the fact that many finance problems are potentially so time-consuming, we will use simplifying formulas for four classes of cash flow streams:
  a. Annuity
  b. Growing Annuity
  c. Perpetuity
  d. Growing Perpetuity

Annuity

- An annuity is a series of equal payments made at regular intervals, or an annuity can be defined as a level stream of regular payments that lasts for a fixed number of periods.
  - E.g. Monthly mortgage payments or pensions received when people retire are often in the form of an annuity, or quarterly payments by a company into an employee retirement account.
- The two basic purposes of an annuity are to:
  1. Accumulate money for a future need such as a cash payment for a house, or
  2. Make regular monthly payments from an accumulated sum of money such as monthly benefits from a retirement plan. Note that the amount of money decreases as payments are made.

- Annuities are among the most common kinds of financial instruments, and the following definitions should be noted:
  1. Ordinary Annuity - One in which payments are made at the end of each period
  2. Payment Period -Time between payments.
  3. Term of the Annuity - Time needed for all payments to be made

- Note that interest calculations for annuities are done using compound interest.
- The total amount in an annuity at a future date is the amount, compound amount, or future value of the annuity, and these three terms will be used interchangeably.

Example

- For example, let's assume person A uses an annuity to save some money to purchase a car. Let's further assume that person A deposits K3000 at the end of each year for 6 years into an account earning 8% compounded annually.
- We know that the first payment made at the end of year 1 earns interest for five years, and note that the last payment does not earn interest because the annuity ends on the day of the last payment i.e. end of year six and thus no interest is paid.
- Thus the future value of the annuity is the sum of the compound amounts of all 6 payments.

Present Value Of An Annuity

- The present value of an annuity is found using the following formula:

$$P.V = \frac{C}{1+r} + \frac{C}{(1+r)^2} + \cdots \ldots .. + \frac{C}{(1+r)^T}$$

- Assuming constant cash flows for T years
- Note that this formula above assumes that the payments are made annually or that the annuity begins at date one.

- After some algebraic manipulations, the present value of an annuity simplifies to:

$$P.V = C\left[\frac{1}{r} - \frac{1}{r(1+r)^T}\right]$$

- where C is the constant cash flow (or periodic payment) for T years

- Note that $\left[\frac{1}{r} - \frac{1}{r(1+r)^T}\right]$ is called the annuity factor.

Example

James has just won the state lottery, paying K50 000 a year for 20 years, and James is to receive his first payment a year from now. The state advertises this as the million-kwacha lottery because K50 000 x 20 = K1 000 000. If the interest rate is 8%, what is the true value of the lottery?

Solution

- The present value of the million kwacha lottery is:

$$P.V = C\left[\frac{1}{r} - \frac{1}{r(1+r)^{T}}\right] = periodic\ payments \times Annuity\ factor$$

Solution continuation…

$$P.V\ =\ 50\,000\left[\frac{1}{0.08}-\frac{1}{0.08(1.08)^{20}}\right]$$

$$PV. = 50\,000\ \times 9.8181\ =\ K490,905$$

- Rather than being overjoyed at winning, James sues the state for misrepresentation and fraud. His legal brief states that he was promised K1 million but received only $K490,905$.

Growing Annuity

- In practice, cash flows in a firm are most likely to grow due either to real growth or to inflation.

- Growing annuity can therefore be defined as a finite number of growing cash flows.

- The present value of a growing annuity can be written as:

$$P.V = C\left[\frac{1}{r-g} - \frac{1}{r-g} \times \left(\frac{1+g}{1+r}\right)^{T}\right]$$

- Where C is the payment to occur at the end of the first period, r is the interest rate, g is the rate of growth per period expressed as a percentage and T is the number of periods for the annuity.

Example

- Mary has just been offered a job at $K80\,000$ a year and expects her salary to increase by 9% a year until her retirement in 40 years. Given an interest rate of 20%, what is the present value of her lifetime salary?

Solution

- We assume that she is paid the $K80\,000$ salary one year from now, and that the salary will continue to be paid in annual installments.

$$P.V = C\left[\frac{1}{r-g} - \frac{1}{r-g} \times \left(\frac{1+g}{1+r}\right)^{T}\right]$$

Solution Continuation..

$$C = 80\,000, r = 20\% = 0.2, g = 9\% = 0.09$$

and

$$T = 40\ years$$

$$P.V = 80000\left[\frac{1}{0.2-0.09} - \frac{1}{0.2-0.09}\left(\frac{1.09}{1.2}\right)^{40}\right] = K711,731$$

Perpetuity

- A perpetuity can be defined as a constant stream of cash flows without end.
- For example, an investor purchasing a bond such as consols (British bonds) may be entitled to receive yearly interest from the government forever.
- Let's suppose a consol pays a coupon of C kwacha each year forever. The present value formula will be given as:

P. V= $\frac{C}{1+r} + \frac{C}{(1+r)2} + \frac{C}{(1+r)3} + \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$

- The above series is called a geometric series, and we can observe that the present value of the consol is just the present value of all its future coupons/payments.

- After some algebraic manipulation of the present value formula, the present value of a bond paying constant cash flows forever becomes

$$P.V = \frac{C}{r}$$

Example

Suppose a bond pays a constant amount of k100 a year in perpetuity and that the interest rate is 8%. What is the value of the bond?

Solution

$$PV = \frac{100}{0.08} = K1,250.$$

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Growing Perpetuity

- Let's suppose the expected cash flows are $K100{,}000$ next year and that these cash flows are expected to rise at 5% per year.
- If we assume that the rise will continue indefinitely, then the cash flow stream is termed a growing perpetuity.
- Algebraically, we can write the present value formula as:

$$\boldsymbol{PV = \frac{C}{1+r} + \frac{C(1+g)}{(1+r)^2} + \frac{C(1+g)^2}{(1+r)^3} + \ldots + \frac{C(1+g)^{N-1}}{(1+r)^N} + \cdots \ldots}$$

- Where C is cash flow to be received in period one, g is the rate of growth per period expressed as a percentage and r is the appropriate interest rate. After some algebraic manipulation, the above formula reduces to:

$$P.V = \frac{C}{r-g}$$

- Assuming that the appropriate discount rate is 11%, the P.V of the cash flow in the above example is:

$$P.V = \frac{C}{r-g} = \frac{100{,}000}{0.11-0.05} = K1,666{,}667$$

- Note the following about the growing perpetuity:

    i. The numerator is the cash flow in period one and not at date 0.

    ii. The interest (r) must be greater than the growth rate (g) for the growing perpetuity formula to work.

    iii. The timing assumption

        - Its assumed that cash flows are received and disbursed at regular and discrete time points.
        - However, cash generally flows into and out of real-world forms both randomly and nearly continuously.
        - Thus the growing perpetuity formula can be applied only if we assume a regular and discrete pattern of cash flow.

Amortization

Amortization

- Amortization is the process of paying off debt over time in regular instalments of interest and principal sufficient to repay the loan in full by its maturity date.
- It is a technique used to periodically lower the book value of a loan or an intangible asset over a set period of time.
- For a loan, amortization focuses on spreading out loan payments over time.

- Amortization is used:
    1. in the process of paying off debt through regular principal and interest payments over time.
    2. also refer to the practice of spreading out capital expenses related to intangible assets over a specific duration—usually over the asset's useful life—for accounting and tax purposes.
- Note that under amortization, a higher percentage of the flat monthly payment goes toward interest early in the loan, but with each subsequent payment, a greater percentage of it goes toward the loan's principal.
- This we shall see very soon.

Procedures regarding computations under Amortization

- Amortization can be calculated using most modern financial calculators, spreadsheet software packages.
- Amortization schedules begin with the outstanding loan balance.
- To arrive at the amount of monthly payments, the interest payment is calculated by multiplying the interest rate by the outstanding loan balance and dividing by 12.
- The amount of principal due in a given month is the total monthly payment (a flat amount) minus the interest payment for that month.

- For the next month, the outstanding loan balance is calculated as the previous month's outstanding balance minus the most recent principal payment.
- The interest payment is once again calculated off the new outstanding balance, and the pattern continues until all principal payments have been made, and the loan balance is zero at the end of the loan term.

Finding the payment using the formula for amortization

- The formula given below is used to find the periodic payment needed at the end of each period to amortize a loan with interest i per period, over n periods. Thus

$$\boldsymbol{Payment = Loan\ amount \times Number\ from\ amortization\ table}$$

Example

Mubanga is a mechanic at a Toyota Zambia dealership. He was so impressed with the quality of vehicles that he purchased an IST at a cost of $K29{,}400$, including tax, title, and license, after the rebate. He made a down payment of $K3500$ and financed the balance at a rate of 6% per year for 4 years. Find

1. the monthly payment,
2. the portion of the first payment that is interest,
3. the balance due after one payment,
4. the interest owed for the second month, and
5. the balance after the second payment.

Solution

(1) Amount financed $= K29{,}400 - K3500 = K25{,}900.$

Use $\frac{6\%}{12} = 0.5\%$ per month and 4 years ×12 = 48 months

From the table 48 and 0.5% yields the value of 0.02349.

Monthly payment therefore $= K25{,}900 \times 0.02349 = K608.39$

(2) Interest for first month $= I = PRT$

$$I = K25{,}900 \times 0.06 \times \frac{1}{12} = K129.50$$

Amount of first payment applied to principal $= K608.39 - K129.50$

$$= K478.89$$

(3) Balance after first payment $= K25{,}900 - K478.89$

$$= K25{,}421.11$$

(4) Interest for second month $= PRT$

$$= K25{,}421.11 \times 0.06 \times \frac{1}{12} = K127.11$$

Amount of 2^{nd} payment applied to principal $= K608.39 - K127.11$

$$= K481.28$$

(5) Balance after 2nd payment $= K25{,}421.11 - K481.28$

$$= K24{,}939.83$$

In summary

Payment Number	Payment	Interest	Applied to principal	Loan Balance After Payment
1	$K608.39$	$K129.5$	$K478.89$	$K25421.11$
2	$K608.39$	$K127.11$	$K481.28$	$K24939.83$

How to set up an amortization schedule

Example

Nthumbuzya needed $K22,300$ for equipment for her car wash. Due to issues with her credit history, the bank required her to put $K5000$ down and agreed to lend the balance at a relatively high 12% compounded quarterly with quarterly payments spread out over two years.

i. Find the quarterly payment, and

ii. show the payments in a table called an amortization schedule.

Solution

(i) The amount financed is $K22,300 - K5000 = K17,300.$

- Find the factor from the table using $2 \times 4 = 8$ quarters and

$\frac{12\%}{4} = 3\%$ per quarter.

From the table 8 and 3% corresponds to the factor of 0.14246

- Then multiply the amount financed by the factor to find the payment.

Payment $= K17,300 \times 0.14246 = K2464.56$

Amortization Schedule

Payment Number	Amount of Payment	Interest for Period	Portion to Principal	Principal at End of Period
0	–	–	–	$K17300.00$
1	$K2464.56$	$K519.00$	$K1945.56$	$K15354.44$
2	$K2464.56$	$K460.63$	$K2003.93$	$K13350.51$
3	$K2464.56$	$K400.52$	$K2064.04$	$K11286.47$
4	$K2464.56$	$K338.59$	$K2125.97$	$K9160.50$
5	$K2464.56$	$K274.82$	$K2189.74$	$K6970.76$
6	$K2464.56$	$K209.12$	$K2255.44$	$K4715.32$
7	$K2464.56$	$K141.46$	$K2323.10$	$K2392.22$
8	$K2463.99$ *	$K71.77$	$K2392.22$	$K0$

Note that the last payment that has been asterisked (*) differs slightly so that the final debt ends up at exactly $K0$.

Note the following about the Amortization Schedule:

- The table illustrates a very important point!
- Notice that interest is high at the time of the first
- payment but eventually goes to nearly zero.
- This is because the loan balance is high at first—the amount of interest depends on the amount owed.

End of Lecture 5